Kejing Xia

kxia39@gatech.edu Homepage Scholar GitHub LinkedIn

Education

Georgia Institute of Technology *Aug. 2025 – Present*
Ph.D. Student in Computer Science
- Advisor: Prof. Wenke Lee

Wuhan University *Sep. 2021 – June 2025*
B.Eng. in Communication Engineering (Artificial Intelligence Track)
- GPA: 3.90/4.00 , Rank: Top 5% , First-Class Academic Scholarship

Research Interests

My research centers on LLM post-training and agentic AI, with particular interests in reasoning, efficient language models, and systems that make language models more useful in interactive environments.
- Explicit and Implicit Reasoning (Latent Reasoning, Persistent Working Memory, RL Post-Training)
- Efficient Language Models (Diffusion LLM Decoding, Long-Context State Space Models)
- Language Model Agents (Computer-Use Agents, Source Learning Agents)

Experience

• **Georgia Institute of Technology, GTISC Lab** | Ph.D. Student *Jan. 2026 – Present*
Advisor: Prof. Wenke Lee
Research Focus: *Reasoning, Agents, and LLM Post-Training*
- **MetaState**: Added a lightweight recurrent working memory around a frozen dLLM to carry continuous information across denoising steps, where standard discrete diffusion models pass only discrete tokens.
- **Rhythm of Thought**: Made latent reasoning aware of its own progress: at each decoding step the model predicts a Gaussian over reasoning progress and turns it into an uncertainty-calibrated phase signal that modulates decoder layers, trained with RL.
- **SourceLearn**: Contributed to a framework that moves LLM agents from accessing a persistent source at query time to learning it, building a reusable source model through self-directed study and task-guided refinement.

• **Zoom Communications, Inc., AI Incubation** | Research Intern *May 2026 – Aug. 2026*
Mentor: Dr. Renqian Luo
Research Focus: *Efficient Computer-Use Agents*
- **LookAgain-CUA**: Instead of re-reading the screen after every GUI action, trained agents to learn observation boundaries and execute several actions per model call. We built BoundaryTrace to turn trajectories into multi-action supervision, then trained with self-refined SFT and cost-aware RL.

• **Georgia Institute of Technology, EIC Lab** | Undergraduate to Ph.D. Student *May 2024 – Dec. 2025*
Advisor & Mentors: Prof. Yingyan (Celine) Lin, Yonggan Fu, Zhifan Ye
Research Focus: *Efficient Diffusion Decoding and Long-Context Modeling*
- **LongMamba**: Found that Mamba's hidden channels split into local and global ones, traced long-context failure to state decay in the global ones, and enlarged their receptive fields with training-free token filtering.
- Also contributed to R²-dLLM (redundancy-aware dLLM decoding), training-free soft token interpolation for masked diffusion LLMs, and LAMB (attention-guided token filtering for SSMs).

• **Bosch Research & Innovation Software Development Co., Ltd.** | Research Intern *July 2024 – May 2025*
Mentor: Dr. Youjian Zhang
Research Focus: *LiDAR-free Urban Scene Reconstruction*
- **D²GS**: To reconstruct urban scenes without LiDAR, co-optimized diffusion-generated dense depth and 3D Gaussian Splatting in an iterative loop, with progressive pruning and road-specific geometric constraints.

• **Wuhan University, Signal Processing Lab** | Undergraduate Researcher *Oct. 2023 – Apr. 2024*
Advisor: Prof. Lei Yu
Research Focus: *3D Vision, Event-based Vision*
- **BRS-E²NeRF**: Combined events with RGB through event-enhanced re-blurring and double-cycle optimization to jointly correct motion blur and rolling-shutter distortion for 3D reconstruction.

Publications

** equal contribution*

[1] **K. Xia**, Z. Du, S. Yao, L. Fu, H. Shen, X. Ma, L. Wei, M. Li, X. Yuan, D. Shi, Q. Jin, B. A. Prakash, W. Lee, and R. Luo. When to Look Again: Learning Observation Boundaries for Efficient Computer-Use Agents. Submitted to ACL ARR 2026.
[2] **K. Xia**, L. Wei, M. Li, D. Shi, X. Yuan, Z. Du, Q. Jin, and W. Lee. Rhythm of Thought: Progress-Aware Latent Reasoning via Reinforcement Learning. Submitted to ACL ARR 2026.
[3] Z. Du, **K. Xia**, J. Zhang, S. Yao, Z. Shi, Z. Zheng, J. Luo, X. Yuan, D. Shi, Q. Jin, and W. Lee. GEvol: Graph-Guided Benchmark Evolution for Terminal Agents. Submitted to ACL ARR 2026.
[4] **K. Xia**, M. Li, L. Wei, Z. Du, X. Yuan, D. Shi, Q. Jin, and W. Lee. MetaState: Persistent Working Memory Enhances Reasoning in Discrete Diffusion Language Models. COLM 2026. [Paper] [Code] [Website]
[5] **K. Xia***, J. Jia*, K. Jin, Y. Bai, L. Sun, D. Tao, and Y. Zhang. D²GS: Dense Depth Regularization for LiDAR-free Urban Scene Reconstruction. NeurIPS 2025. [Paper] [Code] [Website]
[6] Z. Ye*, **K. Xia***, Y. Fu, X. Dong, J. Hong, X. Yuan, S. Diao, J. Kautz, P. Molchanov, and Y. Lin. LongMamba: Enhancing Mamba's Long-Context Capabilities via Training-Free Receptive Field Enlargement. ICLR 2025. [Paper] [Code] [Website]
[7] H. Wang*, **K. Xia***, Y. Wang, C. Zhang, M. Lin, and L. Yu. BRS-E²NeRF: Event Enhanced Neural Radiance Fields from Blurry Rolling Shutter Images. Pattern Recognition, 2026. [Paper]
[8] Q. Jin, L. Tung, K. Li, Q. Shi, Y. She, H. Jia, H. Zhao, **K. Xia**, Z. Du, et al. ChainSWE: Benchmarking Coding Agents on Multi-Bug Software Maintenance. EMNLP Main 2026. [Paper] [Benchmark]
[9] Z. Ye, Z. Wang, **K. Xia**, J. Hong, L. Li, L. Whalen, C. Wan, Y. Fu, Y. Lin, and S. Kundu. LAMB: A Training-Free Method to Enhance the Long-Context Understanding of SSMs via Attention-Guided Token Filtering. ACL Main 2025. [Paper] [Code]
[10] X. Yuan, X. Diao, Z. Liu, C. Zhang, D. Shi, **K. Xia**, Z. Du, S. Yao, S. Vosoughi, and W. Lee. Priority-Preserving Advantage Aggregation for Multi-Reward Policy Optimization. Preprint, 2026.
[11] Z. Du, **K. Xia**, X. Zhong, Y. Fu, N. Oswald, B. Ji, B. Khailany, P. Molchanov, and Y. Lin. R²-dLLM: Accelerating Diffusion Large Language Models via Spatio-Temporal Redundancy Reduction. Preprint, 2026. [Paper] [Code]
[12] L. Fu, **K. Xia**, Y. Wang, Y. Jin, J. He, X. Yang, H. Liu, Y. Yu, H. Jin, Y. Xiao, W. Lee, B. A. Prakash, and H. Wang. From Knowledge Access to Source Learning: Developing Source-Specific Competence. Preprint, 2026. [Paper] [Code] [Website]
[13] M. Li, **K. Xia**, G. Zhang, Z. Wang, G. Tao, S. Pan, J. Zhai, and S. Ma. EDITOR: Effective and Interpretable Prompt Inversion for Text-to-Image Diffusion Models. Preprint, 2025. [Paper]
[14] Z. Du*, J. Luo*, Z. Zheng, X. Yuan, **K. Xia**, D. Shi, Q. Jin, Q. He, S. Zou, Y. Liang, and W. Lee. PACT: Privileged Trace Co-Training for Multi-Turn Tool-Use Agents. Preprint, 2026. [Paper] [Website]
[15] D. Shi, H. Zhu, X. Yuan, W. Zhao, **K. Xia**, W. Xiao, and W. Lee. CopT: Contrastive On-Policy Thinking with Continuous Spaces for General and Agentic Reasoning. Preprint, 2026. [Paper]
[16] Y. Peng, M. Li, **K. Xia**, R. Zhang, and A. Houmansadr. Membership Inference Attacks on Vision-Language-Action Models. Preprint, 2026. [Paper]
[17] **K. Xia***, L. Wei*, and L. Yu. A Spatio-temporal Event Transformer on Versatile Tasks for Human Behavior Analysis. MiGA Workshop at IJCAI 2024. [Paper]

Honors and Awards

- **Top Reviewer**, NeurIPS 2027 Main Conference — *2026*
- **National Second Prize (Team Captain, Fifth Place)**, RoboCup China Open — *2024*
- **First-Class Academic Scholarship**, Wuhan University — *2023 – 2024*
- **"Luojia" Distinguished Exchange Student Scholarship**, Wuhan University — *2023 – 2024*
- **National-Level Project**, Student Innovation & Entrepreneurship Project — *2023 – 2024*
- **National First Prize**, Huazhong Cup Mathematical Modeling Challenge — *2023*
- **Hubei Province First Prize**, China Undergraduate Mathematical Contest in Modeling — *2023*
- **Outstanding Student**, Wuhan University — *2022 – 2023*
- **Merit Student**, Wuhan University — *2021 – 2022, 2023 – 2024*

Teaching and Service

- **Teaching Assistant**, CS 6035 Introduction to Information Security, Georgia Tech — *Spring 2026*
- **Teaching Assistant**, CS 6262 Network Security, Georgia Tech — *Spring 2026*
- **Teaching Assistant**, CS 4803/8803 Efficient Machine Learning, Georgia Tech — *Fall 2025*
- **Conference Reviewer**, ECCV, COLM, NeurIPS, EMNLP, ICLR, NAACL — *From 2026*